BST.java

You will write the Binary Search Tree method remove.

The remove method will exist in the BST class, which will extend the BinarySearchTree2 class. The BinarySearchTree2 class is an abstract class that is missing the method

public BinaryNode remove(BinaryNode startNode, Comparable target)

which you are expected to write.

This method is the helper method for the BinarySearchTree2 method

public BinaryNode remove(Comparable target)

This means, when you write your remove method, you do not have to worry about the root edge case occurring, since the remove(Comparable) method has already checked that.

When writing your remove method, you may consider the following methods have been inherited from the BinarySearchTree2 method:

BinaryNode search(Binary Node startNode, Comparable target)
// this method returns the parent of the target null, or null if target is not in the tree

BinaryNode successor(BinaryNode x)
// returns the inorder successor of the node x

void swap(BinaryNode x, BinaryNode y)
// swaps the Comparable value of nodes x and y

String toString()
// will return a String of the inorder traversal of the BinarySearchTree2

When writing your remove method, you may consider the following methods are in the BinaryNode class:

BinaryNode left()
// returns the left child of the node, null if no child exists

BinaryNode right()
// returns the right child of the node, null if no child exists

Comparable getValue()
// returns the Comparable value of the node

BinaryNode setLeft(BinaryNode x)
//sets the left child to x

BinaryNode setRight(BinaryNode x)
//sets the right child to x

Write the BST method
public BinaryNode remove(BinaryNode startNode, Comparable target)